

Model Evaluation Run

Predicting loan approval with Random Forest

79.2%
TEST ACCURACY

598 applications, 100 decision trees, and a held-out test split of 120. Diagnostic breakdown of model confidence and feature weights.

598

Applications In Dataset

478

Used For Training

120

Held Out For Test

0.774

ROC AUC Score

Prediction Distribution

Comparison between ground truth labels and model predictions on the 120 test cases.



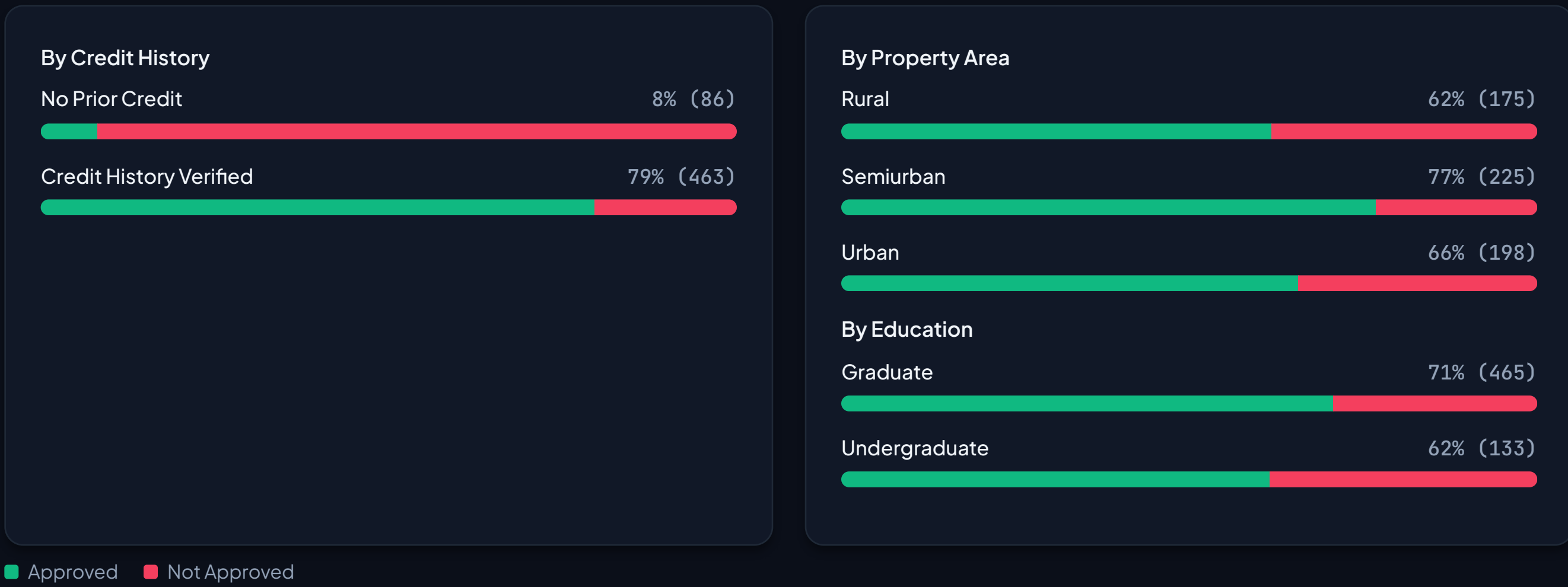
Feature Importance Weights

Gini importance contribution of each input variable toward splitting tree branches.



Baseline Approval Rates by Trait

Raw categorical distributions across the pre-training dataset.



Pre-training Missing Value Imputation

Columns requiring median (numeric) or mode (categorical) replacements.

Dependents 12 null

LoanAmount 21 null

Loan_Amount_Term 14 null

Credit_History 49 null

Model: RandomForestClassifier(n_estimators=100, random_state=42)

80 / 20 Train-Test Split